GURUPRAKASH Y

Generative AI Engineer

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PROFESSIONAL SUMMARY

Generative AI & Python Developer with a strong foundation in Python, SQL, Machine Learning, and Large Language Models (LLMs). Skilled in building AI-powered applications using Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, AWS Bedrock, FastAPI, and Vector Databases through academic projects. Experienced in applying classical machine learning techniques for real-world classification problems such as fraud detection. Passionate about developing intelligent applications and continuously learning emerging AI technologies.

TECHNICAL SKILLS

Programming: Python, SQL
Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, Semantic Search
Frameworks: LangChain, LangGraph, Hugging Face Transformers
LLM Platforms: OpenAI API, AWS Bedrock
Foundation Models: GPT, Claude, Amazon Titan
Backend: FastAPI, Flask, Streamlit
Machine Learning: Scikit-learn, TensorFlow, Logistic Regression, Random Forest, SVM
Data Analysis: Pandas, NumPy
Vector Databases: ChromaDB, FAISS, Pinecone
Cloud: AWS (EC2, S3, Bedrock)
Tools: Git, GitHub, Docker

PROJECTS

AI-Powered Financial Document Q&A System using RAG

Python | LangChain | FAISS | Pinecone | OpenAI API | AWS Bedrock | Streamlit

- Developed a Retrieval-Augmented Generation (RAG) application to answer user queries from financial PDF documents using Python and LangChain.
- Implemented document extraction, recursive/semantic chunking, and embedding generation to convert unstructured documents into searchable vectors.
- Integrated FAISS/Pinecone vector database for semantic similarity search and used OpenAI/AWS Bedrock LLMs to generate context-grounded responses.
- Built an interactive Streamlit interface for document upload and conversational question answering while reducing irrelevant responses through retrieval-based prompting.

Agentic AI Research Assistant using LangGraph

Python | LangChain | LangGraph | LLM Tool Calling | RAG

- Developed an Agentic AI research assistant using LangGraph and LangChain to automate multi-step research and information retrieval tasks.
- Designed an agent workflow involving research, retrieval, validation, and response generation for handling complex user queries.
- Integrated LLM tool calling and RAG to retrieve relevant information and provide context-aware responses.
- Implemented state-based agent orchestration with memory and validation mechanisms to improve response reliability.

Online Payment Fraud Detection System

Python | Scikit-learn | Pandas | NumPy | Machine Learning

- Developed a machine learning system to detect fraudulent online payment transactions, designed for real-time risk assessment.
- Applied data preprocessing and feature engineering to extract meaningful transaction-level patterns and improve model input quality.
- Trained and compared multiple ML models — Logistic Regression, Random Forest, and SVM — to identify the best-performing classifier for fraud detection.

- Enabled dynamic risk scoring to flag suspicious transactions, improving detection accuracy while reducing false positives.
- Built a scalable, modular pipeline architecture suitable for deployment in high-volume environments such as banks and e-commerce platforms.

EDUCATION

Bachelor of Technology – Information Technology **2021 – 2025**
Mahendra Institute of Technology, Namakkal | CGPA: 7.77

CERTIFICATIONS

- Python Bootcamp – Udemy
- Python & Agentic AI Development Intern – Code99 IT Academy

LANGUAGES

English, Tamil, Kannada